

Lecture 19 Scribe

N-Variate Joint PDF / PMF / CDF

CSE400 – Fundamentals of Probability in Computing

Introduction

Topic: N-Variate Joint Distribution

Outline

1. Joint Distribution of N Random Variables
2. Definition
3. Example-1
4. Application: Large MIMO System
5. Joint PMF, PDF, CDF
6. Independence of Random Variables
7. Example-2: Uniformly Distributed 3D Vector
8. Example-3: Trivariate Random Variables

Joint Distribution of N Random Variables

Definition

Let

$$X_1, X_2, X_3, \dots, X_n$$

be n random variables defined on the same probability space.

The joint distribution is:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

Assumptions

- All variables are defined on the same sample space
- Variables may be discrete or continuous
- Joint probabilities satisfy:
 - Non-negativity
 - Total probability equals 1

Example 1: Joint PMF

For two variables X and Y :

$$P(X = x, Y = y)$$

Properties

1. Non-negativity:

$$P(X = x, Y = y) \geq 0$$

2. Normalization:

$$\sum_x \sum_y P(X = x, Y = y) = 1$$

Marginal Distributions

$$P(X = x) = \sum_y P(X = x, Y = y)$$

$$P(Y = y) = \sum_x P(X = x, Y = y)$$

Application: Large MIMO System

Concept

- MIMO = Multiple Input Multiple Output
- Used in advanced wireless systems (e.g., 6G)

Modeling

$$X_1, X_2, \dots, X_n$$

Use of Joint Distribution

- Captures relationships between multiple signals
- Helps in:
 - Signal detection
 - Noise analysis
 - Performance evaluation

Joint PMF, PDF, CDF

1. Joint PMF (Discrete Case)

$$p(x_1, x_2, \dots, x_n) = P(X_1 = x_1, \dots, X_n = x_n)$$

Properties:

- $p \geq 0$
- $\sum p = 1$

2. Joint PDF (Continuous Case)

$$f(x_1, x_2, \dots, x_n)$$

Properties:

$$f(x_1, \dots, x_n) \geq 0$$
$$\int \cdots \int f(x_1, \dots, x_n) dx_1 \dots dx_n = 1$$

3. Joint CDF

$$F(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n)$$

Relation (PDF from CDF)

$$f(x_1, \dots, x_n) = \frac{\partial^n F}{\partial x_1 \partial x_2 \dots \partial x_n}$$

Independence of Random Variables

Definition

Random variables are independent if:

$$p(x_1, \dots, x_n) = p_1(x_1)p_2(x_2) \dots p_n(x_n)$$

or

$$f(x_1, \dots, x_n) = f_1(x_1)f_2(x_2) \dots f_n(x_n)$$

Key Condition

Joint Distribution = Product of Marginals

Implications

- Simplifies calculations
- Removes dependency between variables

Example 2: Uniformly Distributed 3D Vector

Given

$$(X, Y, Z)$$

is uniformly distributed in a region.

Joint PDF

$$f(x, y, z) = \begin{cases} c & \text{inside region} \\ 0 & \text{outside} \end{cases}$$

Finding Constant c

$$\int \int \int f(x, y, z) dx dy dz = 1$$

$$c \cdot \text{Volume} = 1$$

$$c = \frac{1}{\text{Volume}}$$

Interpretation

- Equal probability density inside region
- Zero outside

Example 3: Trivariate Random Variables

Joint PDF

$$f(x, y, z)$$

Marginal PDFs

$$f_X(x) = \int \int f(x, y, z) dy dz$$

$$f_Y(y) = \int \int f(x, y, z) dx dz$$

$$f_Z(z) = \int \int f(x, y, z) dx dy$$

Checking Independence

If:

$$f(x, y, z) = f_X(x)f_Y(y)f_Z(z)$$

then variables are independent.

Summary

- Joint distribution handles multiple random variables
- Types:
 - PMF (discrete)
 - PDF (continuous)
 - CDF (cumulative)
- Marginals obtained using:
 - Summation (discrete)
 - Integration (continuous)
- Independence:
 - Product of marginals